1 This Week's Progress

1.1 General Items

- Contacted Masimo and obtained information regarding data extraction tools and hardware requirements.
- Clarified pricing and licensing details for Trace software and required accessories.
- Continued working with software to make PPG with MAX30102 more sensitive and accurate

1.2 Technical Progress

- Confirmed pricing and requirements for Masimo data extraction setup:
  - Trace software: $300 for the first year, $300/year for renewal
  - Null modem cable: approximately $50
- Learned that Trace offers a two-week trial period
  - It sounds like Hakan was already able to get Trace for two weeks
- Improved circuit performance by tuning LED and sampling parameters:
  - IR LED brightness: 0x2E
  - Red LED brightness: 0x2E
  - Sample averaging: 16
  - Sample rate: 100
  - Pulse width: 215
- Observed that these parameter adjustments resulted in more stable and consistent signal behavior compared to previous configurations, but still run into noise issues and inconsistency.

1.3 Challenges / Notes

- **Current issue:** Sensor readings are still not consistently accurate despite improved signal stability.
- **Impact:** Limits collection of data and slows validation of the system.
- **Plan:** Continue refining sampling parameters and investigate additional factors such as sensor placement and use of a strap/rubber band for constent pressure

2 Weekly Plans

- Begin testing data extraction of Masimo
- Continue optimizing circuit parameters to improve accuracy of readings.